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The impacts of AI conversational agents on EFL learners' oral proficiency and foreign language speaking anxiety

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Foreign language speaking anxiety remains a major affective barrier in EFL oral communication, particularly in classroom speaking activities. This study investigated the effects of an AI-supported speaking intervention on EFL learners' oral proficiency, speaking anxiety and learner's perception across different task types. Adopting a mixed-method quasi-experimental design, 83 Chinese university students participated in an AI-supported speaking intervention, with follow-up interviews conducted with 12 participants. Quantitative results showed that the experimental group outperformed the control group in oral proficiency ($t = 2.81$, $p = 0.006$). No significant difference was found in overall foreign language speaking anxiety between groups. Task-based analyses revealed that AI conversational agent reduced situational speaking anxiety in pair work and presentation tasks ($p < 0.001$), but not in debate or storytelling. Qualitative findings further showed that learners perceived AI conversational agents as providing scaffolded and adaptive support through personalized feedback, prompting, and repetition in a non-judgmental environment, which facilitated oral development and reduced performance-related anxiety. Overall, the findings suggest that AI conversational agents support oral performance and selectively alleviate situational anxiety, while their effectiveness remains contingent on task structure and pedagogical design.

KEYWORDS

AI conversational agents, EFL learning, foreign language speaking anxiety, oral proficiency, ZPD

1 Introduction

Driven by globalization and digitization, English has become a key tool for communication and career development, and speaking ability is regarded as one of the essential skills in the 21st century. But it is also one of the most difficult skills for language learners to master, especially in non-native environments. Speaking a foreign language not only requires linguistic knowledge, such as vocabulary, grammar, and pronunciation knowledge, but also the ability to communicate effectively in real-time conversation (Divekar et al., 2021). But because of the large class, limited classroom time, and lack of authentic situations, many Chinese EFL learners experience stumbling expression, insufficient confidence, and high anxiety (Fathi et al., 2024; Wang et al., 2024; Yuan, 2023). These challenges not only influence learning outcomes, but also weaken the competitive power for learners in academic and future careers.

The development of AI conversational agents offers a solution to these challenges. The natural language processing (NLP) and adaptive learning technologies have enabled chatbots to provide personalized, interactive language practice, evolving from simple scripted agents to sophisticated conversational partners (Rajak et al., 2024). Through collaboration among multiple agents or workflows, such agents can jointly address complex problems (Barbosa et al., 2025). Previous research has showed that the language learning system, such as intelligent tutoring systems and speech recognition, can enhance language acquisition processes by expanding practice space and creating interactive and immersive environments (Divekar et al., 2021; Eswaran et al., 2024; Semyonkina and Pavlova, 2024). AI chatbots have also been found to help speakers improve fluency, expand vocabulary, and provide immediate feedback, thereby improving oral proficiency and reducing anxiety (Araujo and Bol, 2024; Fathi et al., 2024; Muthmainnah, 2024; Yang et al., 2024; Zhou et al., 2025).

However, current studies are mostly based on general-purpose generative chatbots, such as ChatGPT and Cici bot. Although these tools demonstrate broad potential in language learning, they still present several limitations. First, their performance remains highly dependent on prompt design and often struggles with complex or multi-step tasks (Liu et al., 2023), which suggests constraints when applied to classroom-based speaking instruction that requires multi-task collaboration and nuanced decision-making. Second, as they are designed for general users, they lack adaptation to learners' language proficiency, knowledge background, and learning objectives, making it difficult to provide personalized scaffolding across different tasks (Kasneci et al., 2023). Third, their interactions lack sufficient pragmatic awareness and authenticity of communication, which limits their ability to enhance learners' competence in contextualized communication (Nguyen and Ho, 2024). Consequently, their integration and application in classroom teaching remain limited.

Nevertheless, existing research on AI conversational agents has primarily examined foreign language anxiety from a trait-level perspective, but has rarely analyzed its dynamic changes across different speaking tasks (Chen, 2024; Mak, 2011; Woodrow, 2006). Given that task design is closely related to speaking performance (Mora et al., 2023), it is of great significance to understand how agents regulate learners' anxiety in tasks such as pair work, presentations, debates, and storytelling.

To address the gaps mentioned above, this study investigates the impact of AI conversational agents on EFL learners in formal classroom contexts, focusing on oral proficiency and anxiety through tasks of varying complexity. It seeks to uncover the mechanisms of agent support and provide theoretical and practical insights for AI-driven speaking instruction.

2 Literature review

2.1 Oral proficiency and AI conversational agents' intervention

According to Hulstijn (2011), language proficiency can be understood as the degree to which an individual is equipped

with the linguistic knowledge and cognitive resources required to perform effectively in a particular communicative context and modality (e.g., listening, speaking, reading, writing). Based on it, the present study focuses on Second language(L2) oral proficiency, specifically referring to the ability to express thoughts and ideas effectively verbally in L2. It is always measured by a standard speaking proficiency rubric, like TOEFL (Mostafa et al., 2021). In the Chinese EFL context, it is typically measured by the SET-CET rubric, which has three dimensions—accuracy and range, fluency and coherence, relevance and adaptability (Yan and Yang, 2006).

In recent years, AI conversational agents have witnessed remarkable growth in speaking practice. With advances in NLP, machine learning, and speech recognition technologies, chatbots are able to identify key information from text and speech inputs, thereby generating coherent and appropriate responses (Cui et al., 2022). When misunderstandings or unclear expressions occur, they can provide clarification and more accurate feedback, while flexibly introducing learning content and creating opportunities for educational interaction (Kuhail et al., 2022; Labadze et al., 2023).

Research has shown that chatbots enhance EFL learners' oral proficiency. Specifically, chatbots can improve pronunciation by offering real-time feedback that helps learners detect errors and engage in self-correction (Fathi et al., 2024; Tai, 2022). Studies further indicate that learning English with generative AI conversational agents contributes to the development of fluency (Fathi et al., 2024; Tai and Chen, 2023; Zhou et al., 2025), vocabulary (Fathi et al., 2024; Tai, 2022), grammar (Fathi et al., 2024), and communicative flexibility (Zhou et al., 2025).

However, findings also show different results. Some studies have found that chatbots have only limited effects on grammar improvement (Tai, 2022; Zhou et al., 2025), and do not significantly enhance overall speaking performance (Wan and Moorhouse, 2024). Besides that, existing studies have rarely examined oral proficiency dimensions on coherence and appropriateness on topic. Therefore, the design of multi-workflow customized AI conversational agents that can provide authentic contexts and deeper feedback, while further promoting learners' grammatical development, topic appropriateness, and overall speaking performance, remains an important area for future research.

2.2 Foreign language speaking anxiety and AI conversational agents intervention

Foreign language speaking anxiety (FLSA) is a widely recognized affective factor in L2 acquisition, referring to tension, fear, and discomfort when learners are required to speak in a foreign language, particularly in front of peers or teachers (Erdiana et al., 2020). Scholars generally distinguish between trait anxiety, which is relatively stable, and state anxiety, which fluctuates depending on the situation (Gregersen et al., 2014). FLSA is often associated with learners' difficulties with pronunciation and communication, which can heighten concerns about making mistakes and being negatively evaluated, particularly under performance pressure. These

experiences may discourage participation in oral tasks and ultimately limit the development of speaking ability (Cui, 2011; Yang et al., 2024).

With the rise of AI, AI conversational agents have increasingly been employed to address learners' speaking anxiety. Existing research demonstrates that AI-supported interventions can alleviate anxiety by providing non-judgmental conversational partners in which learners can practice autonomously, receiving immediate corrective feedback, thereby enhancing learners' confidence and willingness to communicate (Chen, 2024; Muthmainnah, 2024; Xiong et al., 2024). For instance, Moussalli and Cardoso (2020) observed that interactions with Amazon Echo created a less intimidating communicative atmosphere, which helped reduce learners' emotional barriers. Similarly, Chen (2024) found a moderate negative correlation between ChatGPT usage frequency and FLSA among university students in Macao, indicating that more frequent use of ChatGPT was associated with lower anxiety levels.

Teaching intervention studies further support these findings. Muthmainnah (2024), in a mixed-methods design, examined the effects of practicing with the AI CiciBot and reported significant improvements in learners' oral proficiency and self-confidence, along with a reduction in anxiety. Likewise, Kim and Su (2024) and Yuan (2023) highlighted that real-time corrective feedback from chatbots reduces learners' fear of making mistakes in front of peers or teachers, thereby fostering confidence and motivation. Ismail and Hastings (2021) also showed that animated chatbots providing motivational and explanatory feedback could significantly reduce learners' anxiety, with gender and performance playing mediating roles in the effectiveness of such interventions.

However, findings across studies are not fully consistent. Ballıdağ and Aydın (2025) found that AI-assisted practice did not reduce anxiety more effectively than traditional peer interaction, suggesting that contextual and task-related factors may influence outcomes. Moreover, current research still has several limitations. First, most studies adopt short-term designs, with limited longitudinal investigation into how learners' speaking anxiety evolves dynamically during sustained interventions and how these changes relate to learners' oral proficiency development over time. Second, previous studies have primarily employed speaking tasks, such as dialogues, with little exploration of how AI tools may affect learners' anxiety across tasks of varying complexity and openness.

In sum, although existing research suggests that AI conversational agents hold promise in reducing speaking anxiety and promoting oral proficiency, gaps remain in classroom-based, longitudinal, and multi-task comparative contexts. Addressing these gaps is essential for advancing the pedagogical integration of AI in EFL speaking instruction.

To address the gaps mentioned above, this study set out to answer the following research questions:

RQ1: How do AI conversational agents impact oral proficiency among EFL students?

RQ2: To what extent do AI conversational agents reduce FLSA among EFL students across different speaking task types?

RQ3: How do EFL students perceive the effectiveness of AI conversational agents?

3 Theoretical framework

The Zone of Proximal Development (ZPD) provides a perspective about how AI chatbot used in spoken class to facilitate language learning. Rather than focusing solely on what learners can accomplish independently, ZPD emphasizes the role of social interaction and mediated support in promoting cognitive development (Vygotsky, 1978). Chaiklin (2003) summarized three assumptions underlying ZPD: generality, assistance, and potential assumption. Teachers or more capable peers are assumed to exert a positive influence on learners' development.

In second language learning, ZPD has often been used to explain how scaffolded interaction supports learners' development. Through feedback, prompts, and guided dialogue, teachers or peers can help learners notice gaps in their language use and gradually extend their communicative abilities. With the development of educational technology, recent studies suggest that AI systems may also function as mediational tools that provide scaffolded assistance during learning. For example, Looi and Jia (2025) identified four ZPD-based levels of personalized learning: personalized responses, step-by-step scaffolding, continuity of conversational memory, and targeted diagnosis with feedback. Al-Hamdani and Yousif (2025) proposed a hybrid "AI+teacher" ZPD model, in which teachers provide contextual and emotional support while AI offers real-time feedback and individualized learning pathways.

Guided by this perspective, the AI conversational agent used in this study was designed to provide scaffolded support during students' speaking practice. The agent offered corrective feedback on grammar and pronunciation, helping learners notice and repair errors. When learners struggled to continue a conversation, the system generated hints and guiding prompts to support dialogue completion. The agent also functioned as a collaborative interaction partner in oral activities such as role-play and debate, sustaining interaction when peer communication became limited.

In addition, the AI agent provided encouragement and positive feedback to create a non-judgmental environment that may reduce speaking anxiety and promote active participation. In this study, ZPD therefore serves as a conceptual framework guiding the design of AI-mediated speaking activities, in which adaptive feedback and prompts support learners in performing tasks within their potential developmental level (see Figure 1). Based on the ZPD framework, this study conceptualizes the AI conversational agent as a scaffolded interaction partner that supports learners' spoken language development through feedback, prompts, and sustained dialogue.

4 Methodology

4.1 Research design

This study employs an explanatory sequential mixed methods design. In the quantitative section, a quasi-experimental design with an instructional intervention is adopted. To examine the function of AI conversational agents in a real classroom context, an empirical teaching setting is established to test the

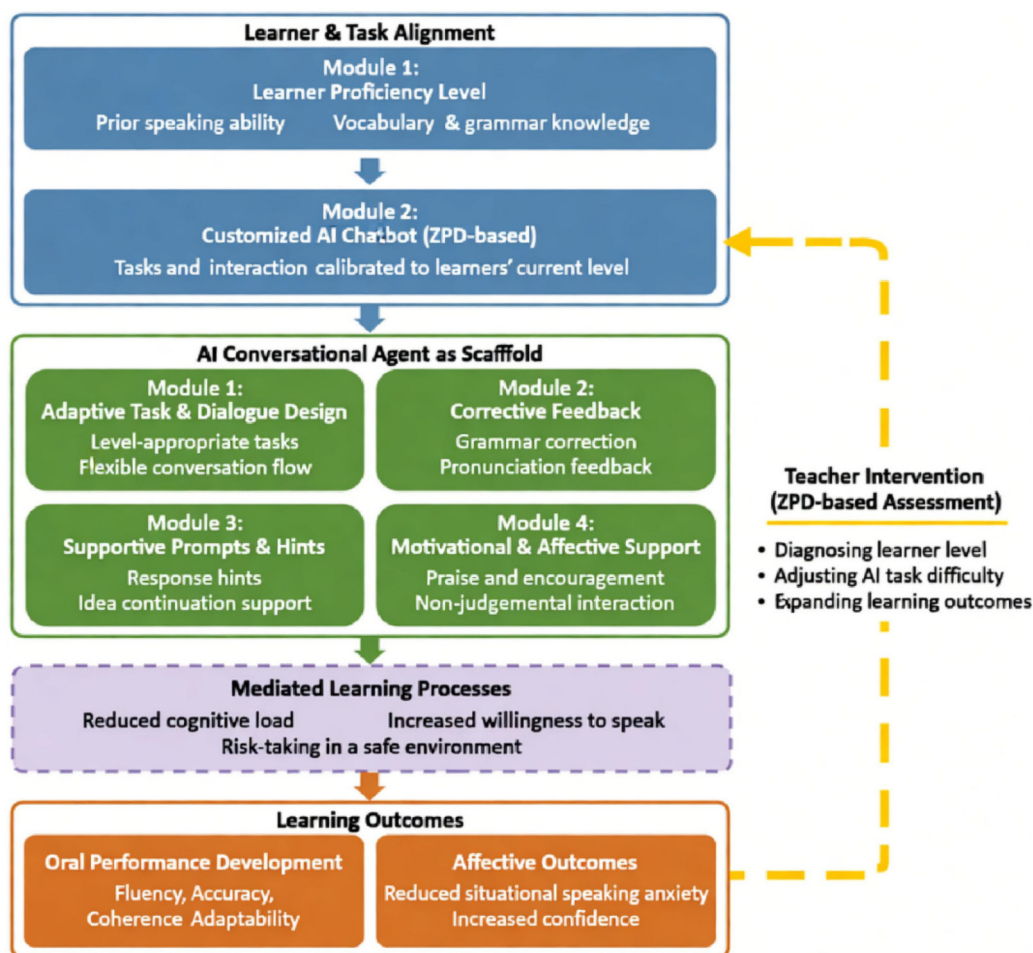


FIGURE 1
ZPD-based framework of customized AI conversational agents for EFL speaking development.

intervention. Creswell and Guetterman (2019) note that experimental research is appropriate when the researcher aims to establish a possible cause-and-effect relationship between independent and dependent variables, and quasi-experimental approaches provide a feasible alternative in educational contexts. Also, semi-structured interviews are conducted to further interpret and explain the quantitative findings, providing deeper insights into participants' experiences and perceptions. This mixed methods approach allows for both generalizability and contextualization (Mertens, 2019), thereby enhancing the validity and richness of the research outcomes.

4.2 Participants

In this study, cluster random sampling was employed to select two intact sophomore English classes as participants for the teaching intervention, with class one ($n = 42$) serving as the control group (CG) receiving traditional interaction and class two ($n = 41$) as the experimental group (EG) interacting with AI conversational agents. According to Mertens (2019), this meets the minimum sample size requirement for a quasi-experimental design. Most students had at least 12 years of English learning experience and demonstrated an intermediate proficiency level.

Both classes were taught by the same instructor, who had over 12 years of EFL teaching experience and an intermediate level of digital literacy.

In the qualitative phase, purposive sampling was used to select 12 students with different language proficiency from the EG for semi-structured interviews. This number aligns with Guest et al. (2006) recommendation that data saturation is typically reached within 12 interviews, while also considering proficiency as a key factor influencing AI interaction (Shikun et al., 2024).

4.3 Materials and instruments

The instruments used in this study include an AI conversational agent, a scale, a questionnaire, an oral proficiency test, and an interview guide.

The AI Conversational Agent was designed by the researcher for this study called "Cindy HUANG" based on the Coze Platform and Cicibot. Coze is a no-code AI agent development platform that allows researchers to design customized conversational agents through structured prompts, workflow configuration, and knowledge integration. It enables developers to define conversation logic, prompt templates, and response strategies without extensive programming. The platform was

selected for this study because it allows flexible prompt engineering and workflow design, which made it possible to implement pedagogically informed interaction strategies such as scaffolding and Socratic questioning.

This agent is customized for college students to practice English. It can interact with students, understand their requirements, and provide corresponding answers. The customization showcase is as follows:

- (a) The avatar-like chatbot uses the instructor's name, image, and even voice, to arouse intimacy with the students to simulate their engagement.
- (b) The prompts used to design the agent are preset to match learners' language proficiency levels, so the generated response will not give the learner too heavy "Cognitive load". According to the ZPD theory, the maximal acquisition occurs when the learners are in their ZPD. The prompts embedded in the agent define the expected language proficiency level and guide the model to produce responses slightly above the learner's demonstrated performance. During interaction, the agent analyzes learners' responses (e.g., lexical complexity, grammatical accuracy, and response length) and adjusts its follow-up questions accordingly. Also, the agent provides graduated scaffolding during interaction. When learners produce incomplete or

incorrect responses, the system follows a structured feedback sequence: (1) clarification request, (2) hint provision, and (3) model reformulation if needed. This branching logic was embedded in the prompt design (See [Appendix 1](#)).

- (c) The conversation agent is designed to apply "Socrates questioning" to stimulate students to reflect on their answers and snowball the conversation.
- (d) The preset prompts also correct learners' pronunciation by repeating the error sentences without directly pointing out mistakes. For grammatical or semantic errors, feedback is provided in text form to avoid excessive cognitive load during conversation. To reduce potential frustration, the agent highlights one mistake every two or three conversational turns.
- (e) During the conversation, the agent will encourage or praise the participants at times to create a positive environment. The AI conversational agent is shown in [Figure 2](#) and the examples of different scaffolds are shown in [Figure 3](#).

The Short-Foreign Language Classroom Anxiety Scale (S-FLCSA), developed by [Botes et al. \(2022\)](#), was adopted. This scale consists of eight items rated on a five-point Likert scale ranging from "1 (Strongly Disagree)" to "5 (Strongly Agree)," designed to measure learners' anxiety in an EFL context. In the

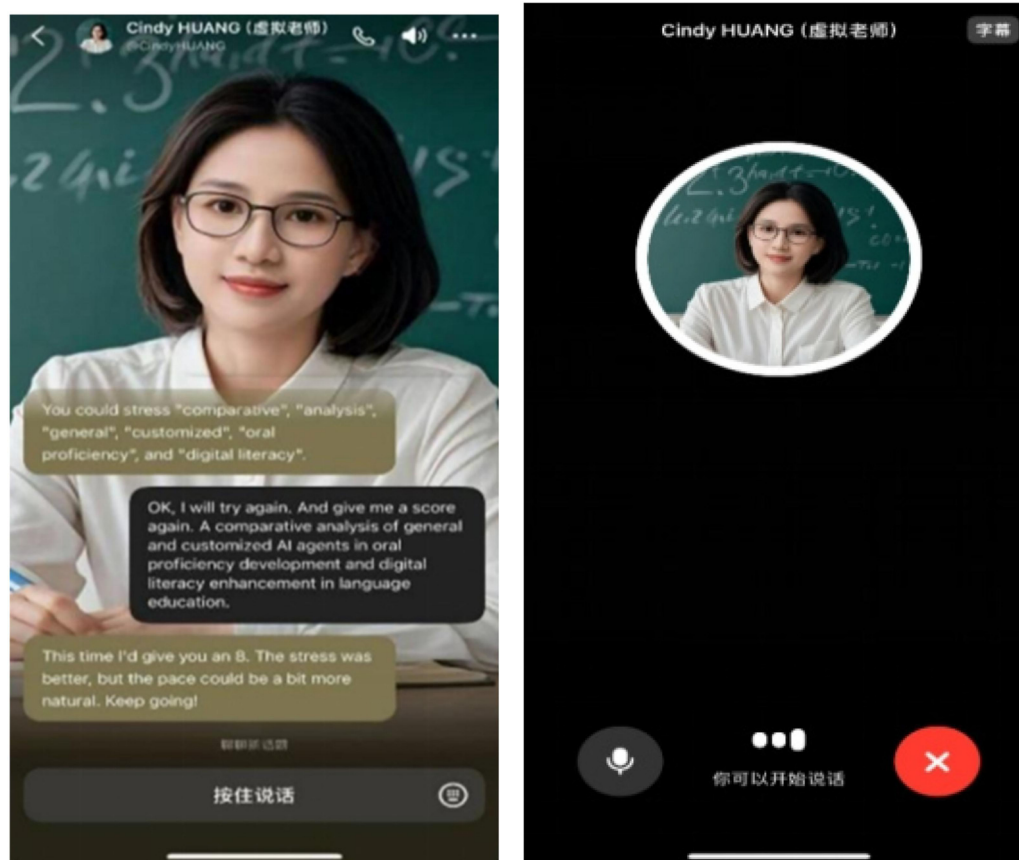


FIGURE 2
Screenshot of AI conversational agent Cindy Huang.



FIGURE 3
Screenshot of different scaffolds.

present study, Cronbach's α was 0.82 in the pre-test and 0.83 in the post-test.

In addition, a Self-Evaluation of Oral Activity Performance questionnaire using a 1–10 scale was administered after each oral activity, in which participants evaluated their speaking anxiety and oral performance. Three experts in educational technology, applied linguistics, and SLA were invited to fill out the Index of Objective Consistency (IOC) independently to evaluate the content validity of survey items. In this study, the IOC analysis result of the Self-Evaluation of Oral Activity Performance Scale is 1.00, indicating strong content validity.

For the oral proficiency test, the standardized CET-SET4 was employed, which is widely used in the Chinese EFL context. The test evaluates accuracy and range, coherence and fluency, as well as relatedness and adaptability, with a total score from 0 to 15 points. Two parallel tests with different content were used for the pre- and post-tests to ensure fairness. The Item-Objective Congruence (IOC) analysis of the test ranged from 0.67 to 1.00, indicating sufficient content validity. Inter-rater reliability was assessed using the Intraclass Correlation Coefficient (ICC), based on a two-way mixed effects model with a consistency definition. All oral responses were audio-recorded and anonymized prior to scoring. To minimize potential expectancy bias, the raters were blind to both the participants' group (experimental vs. control) and the test timing (pre-test vs. post-test). The results showed an ICC of 0.96 (95% CI = 0.94–0.98) for the pre-test and 0.92 (95% CI = 0.88–0.95) for the post-test, suggesting a high degree of agreement between the two raters.

Finally, a semi-structured interview guide was used to gain deeper insights into participants' perceptions of the effectiveness of integrating AI conversational agents into classroom oral activities. The IOC analysis of the interview guide ranged from 0.67 to 1, confirming sufficient content validity of the items.

4.4 Research procedure

The study lasted 16 weeks and included three main stages: pre-test, classroom intervention, and post-test (see Figure 4). In the first week, both the EG ($n = 41$) and the CG ($n = 42$) were introduced to the course. The EG also received short training on the use of AI conversational agents. After that, both groups took the pre-test on oral proficiency and speaking anxiety.

From Weeks 2 to 13, classroom activities followed a regular cycle. Each session included a warm-up, preparation, speaking practice, and review. The difference lay in task support: the CG interacted with peers or the teacher, while the EG interacted with *Cindy HUANG* (an AI conversational agent). Students were required to complete weekly speaking tasks using the AI agent, approximately 120 min per week. The Coze platform automatically recorded duration. The speaking tasks included presentations, pair work, debates, and storytelling, giving students diverse practice opportunities.

In Week 14, both groups completed the post-test of oral proficiency and speaking anxiety. In the final two weeks, semi-structured interviews were conducted with 12 students from the EG to further explore their views on the use of AI conversational agents in class.

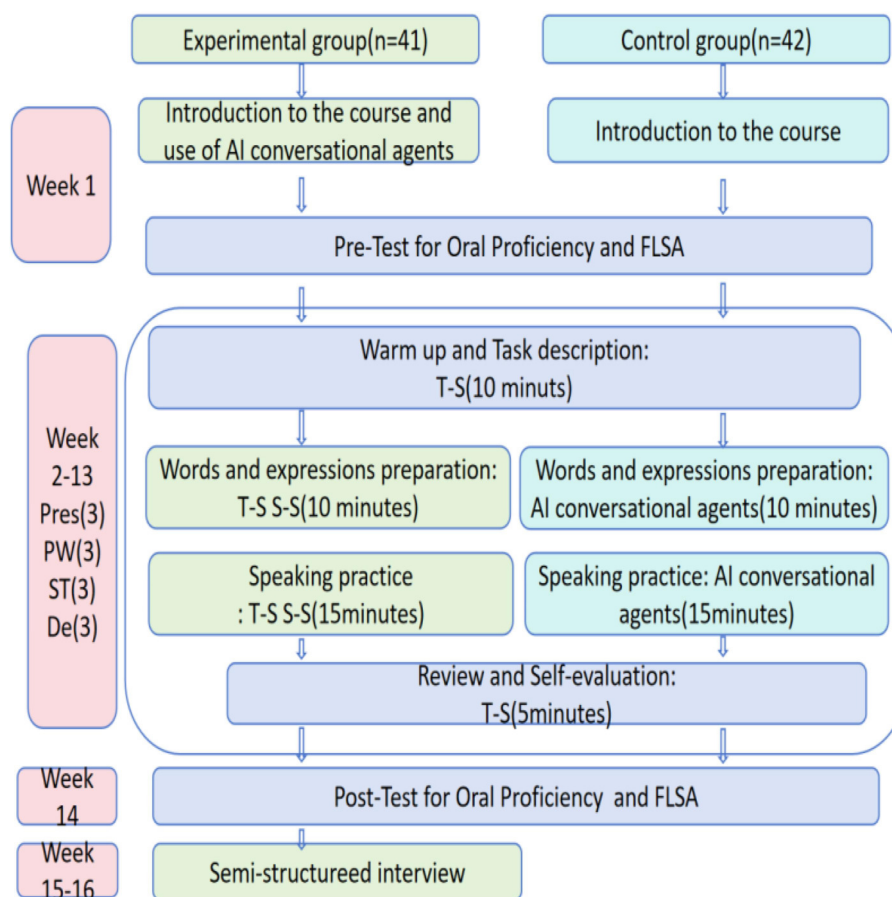


FIGURE 4

Experimental flow chart. T, teacher; S, students; Pres, presentation; PW, pair work; ST, storytelling; De, debate.

4.5 Data analysis

Quantitative analyses were conducted with SPSS 26.0. Descriptive statistics were first calculated for oral proficiency (accuracy and range, coherence and fluency, relatedness and adaptability) and FLSA. For **RQ1**, independent-samples *t*-tests checked baseline equivalence, paired-samples *t*-tests examined within-group changes, and post-test differences between the EG and CG were analyzed with independent-samples *t*-tests. For **RQ2**, the same procedures were applied to FLSA, with an additional mixed-design ANOVA to explore task-specific effects (pair work, debate, presentation, storytelling), followed by Bonferroni-adjusted comparisons when necessary.

Effect sizes were reported to supplement significance testing. Specifically, Cohen's *d* was computed based on the mean difference divided by the standard deviation of the difference scores in paired samples, which is recommended for repeated-measures designs (Cohen et al., 2002). Cohen's *d* values of 0.2, 0.5, and 0.8 were interpreted as small, medium, and large (Cohen et al., 2002), while partial η^2 values of 0.01, 0.06, and 0.14 indicated small, medium, and large effects (Cohen, 2013). The significance level was set at $p < 0.05$.

For **RQ3**, qualitative data from semi-structured interviews with 12 EG participants and structured feedback forms were analyzed through thematic analysis. Transcripts were coded in NVivo by two independent researchers. Both researchers first

conducted open coding to identify initial categories, then met to compare codes. After consensus was reached, axial coding was carried out to group related codes into broader themes. Inter-coder reliability was calculated using Cohen's Kappa, a statistical measure that evaluates the level of agreement between coders while accounting for chance agreement, which exceeded 0.80, indicating strong agreement and ensuring the reliability of the analysis (Creswell and Guetterman, 2019).

4.6 Ethical consideration

Before the study, the research proposal was reviewed and approved by the institutional review board (IRB) (Certificate Number: IRBCMRU 2025/323.23.06). This research meets the ethical guidelines and the protection of the participating learners' rights. Before the research, informed consent was obtained from all the participants, and their privacy and confidentiality were maintained.

5 Results

5.1 Oral proficiency

Independent-samples *t*-tests confirmed no significant differences between the EG and CG at baseline in pre-test oral

performance ($p > 0.05$, $d = 0.15$). Within-group comparisons revealed no significant gains for the CG across the three oral performance dimensions (Cohen's $d = 0.07$ – 0.23). In contrast, the EG demonstrated moderate improvements from pre- to post-test in three dimensions of oral proficiency, accuracy and range ($t = -3.61$, $p = 0.001$, $d = 0.56$) with a gain of 0.27 points, coherence and fluency ($t = -6.33$, $p < 0.001$, $d = 0.99$) with a gain of 0.49 points, and relatedness and adaptability ($t = -7.59$, $p < 0.001$, $d = 1.19$) with a gain of 0.67 (See Table 1). Although this change may appear moderate, it suggests a meaningful improvement for students to respond to interlocutors and adapt to conversational contexts. Post-test comparisons showed the EG significantly outperformed the CG on overall oral proficiency [$t(81) = 2.81$, $p = 0.006$] and on relatedness and adaptability [$t(81) = 5.18$, $p < 0.001$] (See Table 2). These results indicate that AI conversational agents enhanced oral proficiency, particularly in relatedness and adaptability.

5.2 Foreign language speaking anxiety

At baseline, no significant difference was found between EG and CG in pre-test FLSA scores ($t = -0.68$, $p = 0.50$). Paired-samples t -tests revealed that the EG experienced a significant reduction in anxiety from pre-test ($M = 24.95$, $SD = 4.78$) to post-test ($M = 23.63$, $SD = 4.10$), $t = 2.81$, $p = 0.008$, $d = 0.44$, while no significant change was found in the CG ($p > 0.05$) (See Table 3). Post-test comparisons between groups showed lower anxiety in the EG ($M = 23.63$) than in the CG ($M = 25.19$), though the difference did not reach statistical significance ($t = -1.50$, $p = 0.14$) (See Table 4). A mixed-design ANOVA indicated significant main effects of activity type [Wilks' $\Lambda = .839$, $F(3,79) = 5.04$, $p = 0.003$, $\eta^2 = 0.16$]. Mauchly's test of sphericity was significant ($p = 0.001$), indicating violation of the sphericity assumption, so Greenhouse-Geisser corrected results were reported in subsequent analyses. The results showed a

TABLE 1 Results of paired-samples t -tests for Pre- and post-test within groups.

Group	Item	Pretest $M \pm SD$	Posttest $M \pm SD$	t	p	Mean diff	Cohen's d	95%CI
EG $n = 41$	Oral test Score	8.84 ± 2.30	10.27 ± 1.47	-6.99	<0.001	-1.43	1.09	$[-1.84, -1.02]$
	Score 1	3.05 ± 0.80	3.32 ± 0.53	-3.61	0.001	-0.27	0.56	$[-0.42, -0.12]$
	Score 2	2.93 ± 0.80	3.42 ± 0.57	-6.33	<0.001	-0.49	0.99	$[-0.65, -0.33]$
	Score 3	2.87 ± 0.80	3.54 ± 0.49	-7.59	<0.001	-0.67	1.19	$[-0.85, -0.49]$
CG $n = 42$	Oral test Score	9.17 ± 1.99	9.37 ± 1.44	-1.15	0.26	-0.20	0.18	$[-0.55, 0.15]$
	Score 1	3.05 ± 0.71	3.16 ± 0.51	-1.14	0.26	-0.11	0.18	$[-0.31, 0.09]$
	Score 2	3.27 ± 0.65	3.30 ± 0.54	-0.42	0.68	-0.02	0.07	$[-0.16, 0.12]$
	Score 3	2.83 ± 0.74	2.94 ± 0.55	-1.46	0.15	-0.11	0.23	$[-0.26, 0.04]$

Score 1 for accuracy and range, score 2 for coherence and fluency, and score 3 for relatedness and adaptability.

TABLE 2 Results of independent sample t -tests between groups after intervention.

Item	EG($n = 41$) $M \pm SD$	CG($n = 42$) $M \pm SD$	t	p	95% CI
Post-Score	10.27 ± 1.47	9.37 ± 1.44	2.81	0.006	$[0.26, 1.54]$
post_score 1	3.32 ± 0.53	3.16 ± 0.51	1.42	0.16	$[-0.06, 0.38]$
post_score 2	3.42 ± 0.57	3.30 ± 0.54	0.96	0.34	$[-0.13, 0.37]$
post_score 3	3.54 ± 0.49	2.94 ± 0.55	5.18	<0.001	$[0.37, 0.83]$

score 1 for accuracy and range, score 2 for coherence and fluency, and score 3 for relatedness and adaptability.

TABLE 3 Results of paired-samples t -tests for pre- and post-test within groups.

Group	Item	Pretest $M \pm SD$	Posttest $M \pm SD$	t	p	Mean diff	Std. Deviation	Cohen's d	95% CI
EG ($n = 41$)	Anxiety	24.95 ± 4.78	23.63 ± 4.10	2.81	0.008	1.32	3.00	0.44	$[0.37, 2.27]$
CG ($n = 42$)	Anxiety	25.71 ± 5.47	25.19 ± 5.32	0.77	0.45	0.52	4.41	0.12	$[-0.84, 1.88]$

TABLE 4 Results of independent sample t-tests between groups after intervention.

Item	EG(<i>n</i> = 41) M ± SD	CG(<i>n</i> = 42) M ± SD	<i>t</i>	<i>p</i>	95%CI
Post_Anxiety	23.63 ± 4.10	25.19 ± 5.32	−1.50	0.14	[−3.63, 0.51]

TABLE 5 The result of ANOVA.

Source	df1	df2	<i>F</i>	<i>p</i>	Partial η^2
Group	1	81	9.40	0.03	0.10
Activity	2.60	210.43	3.18	0.03	0.04
Group*Activity	2.60	210.43	5.82	0.001	0.07

significant main effect of activity type [$F(2.60, 210.43) = 5.82$, $p = 0.001$, partial $\eta^2 = 0.07$] (See Table 5 and Figure 5), indicating that students' anxiety levels differed significantly across the four types of speaking activities. *post-hoc* analyses showed significantly lower anxiety in the EG for pair work ($t = -5.27$, $p < 0.001$) and presentation ($t = -3.72$, $p < 0.001$), but not for debate or storytelling (See Table 6).

5.3 Effectiveness of AI conversational agents

The analysis of interviews and feedback highlighted four major themes: perceived improvement in oral skills, anxiety reduction, scaffolding support and technical challenges.

5.3.1 Perceived improvement in oral skills

Almost all participants reported gains in fluency, pronunciation, and confidence, as well as the transfer of practiced expressions to real-life communication. S06 reflected, "I noticed I speak more continuously now. I don't pause every few words like before." Similarly, S01 explained, "I used to think a lot before every sentence. Now I can finish them in one go." Pronunciation gains were also noted; S08 said, "When I talk with the agent, it also uses standard English to reply... little by little, not only does my listening improve, but also my pronunciation." Increased confidence was a recurring theme, with S04 sharing, "Now I speak up more often in class. I don't worry about mistakes that much." 8 out of 12 participants students also described direct transfer, such as S10: "During my presentation, I reused some sentence patterns. It helped me stay organized."

5.3.2 Anxiety reduction

Students emphasized that AI conversational agents created a non-judgmental and emotionally safe environment. As S01 put it, "I didn't feel nervous talking to the chatbot. It didn't judge or interrupt me." S06 echoed, "I usually avoid speaking in class, but with the agent, it felt like private practice." Repetition without

social pressure was also valued. S11 explained, "I could try the same sentence or passage several times until I am fluent." Feedback further reduced stress and encouraged persistence. S05 noted, "It helps me notice my grammar mistakes right away. I can correct some expressions that I always neglect."

5.3.3 Scaffolding support

Students recognized that AI conversational agents provided timely prompts, guided feedback, and strategic support. S01 shared, "It said something like 'you could say...', and that helped me to keep on the track." Others stressed the benefit of constructive correction; S07 recalled, "It did not just say 'You can't say that.' It also told me I used the wrong tense and gave me the correct one." Strategic scaffolding also helped students prepare for structured speech. S11 mentioned, "It told me to explain my opinion first, then give reasons. This helped me organize my answer." S05 recalled: "In the debate, I offer defensive opinions and supportive details I have practiced with the agent before." Besides that, students also notice the specialty of customizable AI agent. S07 mentioned, "This AI was different from the chatbots I used before. It understood my level and didn't give me answers that were too difficult or irrelevant to the course".

5.3.4 Perceived cognitive and emotional challenges

8 out of 12 participants also mentioned the limitations in use, mainly focusing on the recognition errors and inconsistent feedback. S01 stated: "I speak very clearly, but sometimes it misunderstands what I say so our interaction will stop." S05 also added: "Sometimes it can't interpret what I say in the right direction, so I had to repeat it and feel frustrated to talk with her." S02 said "When I want to change another topic, AI always wants to follow the same topic." Another challenge is for the lengthy feedback or answer. S06 mentioned "The feedback generated from AI are too long, and it costs me too much time to understand and sometimes I also need to translate into Chinese to understand it." 7 out of 12 participants feel communication is not so authentic. S09 noted that: "When AI praised me, I don't feel encouraged, because I know it is a tool not a real person. But I agree with her evaluation and scoring."

Overall, although there existed some technical challenges or cognitive load, these accounts suggest that students perceived AI conversational agents as supportive partners, instructive teachers and affective friends that not only enhanced their oral proficiency but also created a safe environment for practice and offered scaffolding that could be directly applied to communicative tasks.

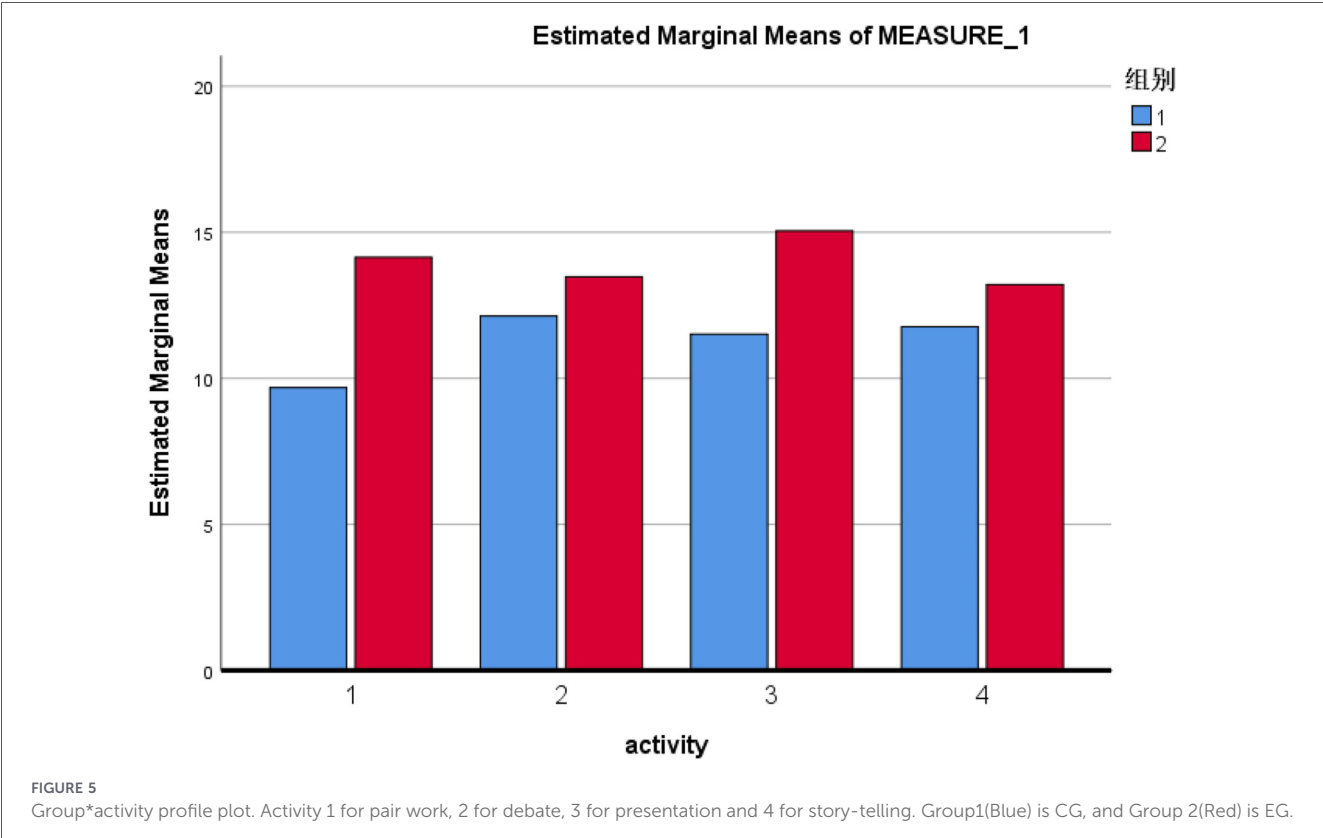


TABLE 6 Result of independent samples t-tests between activities.

Group	Pair work	Debate	Presentation	Story-telling
	M ± SD	M ± SD	M ± SD	M ± SD
EG (n = 41)	9.68 ± 3.28	12.13 ± 4.33	11.51 ± 4.30	11.77 ± 4.71
CG (n = 42)	14.14 ± 4.34	13.48 ± 5.99	15.05 ± 4.35	13.21 ± 5.95
t	−5.27	−1.17	−3.72	−1.23
p	<0.001	0.25	<0.001	0.22

6 Discussion

6.1 The affordance of AI conversational agents in oral performance

The findings indicate that AI conversational agents exerted a positive influence on EFL learners’ spoken performance, particularly in vocabulary, fluency and coherence, and adaptability. Learners reported that interaction with the agents enriched their lexical repertoire, improved grammatical accuracy, and encouraged the use of more complex sentences. Drawing on Vygotsky’s ZPD theory, scaffolded input and adaptive feedback slightly beyond learners’ current proficiency promoted the acquisition of advanced vocabulary and syntactic patterns. Corrective feedback also refined word choice and accuracy, while diverse lexical input activated prior knowledge and expanded expression. These findings align with earlier studies highlighting chatbot affordances for improving linguistic

accuracy and range (Busso and Sanchez, 2024; Qiao and Zhao, 2023; Tai, 2022; Yuan, 2023).

In terms of fluency and coherence, both oral tests and interview data showed that AI-assisted practice significantly reduces students’ hesitation, enhances their fluency, and improves their ability to organize their arguments logically. Some students noted that rehearsing with chatbots helped them transfer expressions into peer discussions and presentations. This reflects ZPD, as it views language learning as adaptive interaction between learners and their environment (Makhdoumi et al., 2017). Unlike earlier rule-based chatbots, generative AI conversational agents provided adaptive feedback across proficiency levels, supporting coherence and consistency in oral performance (Annamalai et al., 2023; Okonkwo and Ade-Ibijola, 2021).

Learners also highlighted adaptability in speech, particularly in maintaining topics and employing strategies such as clarification or elaboration. The AI’s memory and prompting

functions facilitated sustained dialogue and divergent thinking, echoing findings from Pratiwi et al. (2024) on the value of personalized chatbot responses. However, some participants reported mechanical topic shifts and limited naturalness, consistent with concerns raised in previous research (Goh et al., 2024). Also, qualitative research showed that AI conversational agents supported grammatical and discourse competence; they were less effective in promoting spontaneous strategic competence.

In sum, AI conversational agents facilitated improvements in accuracy, fluency, coherence, and adaptability through scaffolded interaction, adaptive feedback, and safe practice opportunities. At the same time, mechanical responses and limited authenticity suggest the need for further refinement to fully support oral competence.

6.2 The affordance of AI conversational agents in anxiety reduction

The study revealed limited effects of AI conversational agents on reducing overall trait-related FLSA. Post-test scores showed no statistically significant difference between groups, diverging from earlier findings that reported substantial anxiety reduction and increased willingness to communicate through chatbots (Muthmainnah, 2024; Zhang et al., 2024). This may be explained by the relatively stable nature of FLSA, which combines trait-like and state-like components, making short-term interventions less likely to shift overall FLSA.

Nevertheless, qualitative data suggested that learners felt less nervous and more comfortable when interacting with AI. Participants described the agents as creating non-judgmental, low-pressure environments, offering multimodal responses and adaptive feedback to cognitive load that may help reduce momentary, task-specific speaking anxiety during interaction, particularly for lower-level learners. Such design features are consistent with prior research emphasizing personalization, adaptive feedback, and user-friendly interfaces as factors lowering speaking anxiety (Nguyen, 2024). Although students sometimes found long corrective outputs tedious, they still viewed AI feedback as less threatening than peer or teacher evaluation (Zou et al., 2023). Automated scoring was also perceived as objective and motivating rather than discouraging, highlighting the potential of AI-based assessment to support a less anxiety-provoking interactional environment. These affordances posit that low-pressure conditions facilitate greater language uptake.

Task-based analysis revealed variation. A reduction in task-specific, state anxiety was evident in pair work and presentation activities, where AI acted as a supportive partner, scaffolding idea organization and lowering fear of social judgment. These results resonate with the ZPD, where adaptive support promotes confidence and performance (Al-Hamdani and Yousif, 2025). By contrast, no significant decline was observed in debates or storytelling. Learners reported that the high cognitive demands of debates heightened anxiety, while storytelling tasks offered limited scope for AI scaffolding. This pattern may be explained from a Task-Based Language Teaching (TBLT) perspective, which suggests that different tasks place different cognitive

demands on learners and draw on their limited attentional resources (Ellis et al., 2019). The possible reason is that narrative skills rely more on personal creativity and open-ended meaning construction, which may reduce the immediate usefulness of AI-generated prompts or feedback during task performance. Similarly, debate tasks often require learners to generate arguments and respond quickly during interaction, which may increase cognitive load and leave less capacity to process AI support. This also suggests that AI primarily alleviates state anxiety in structured or scaffolded contexts, but has limited impact on trait anxiety (Mora et al., 2023; Vytal et al., 2012).

In sum, while AI conversational agents did not significantly reduce overall trait-level FLSA, they created supportive environments that lowered situational anxiety in specific tasks, particularly pair work and presentations. These results highlight a task-specific affective benefit rather than an overall reduction in trait anxiety, highlighting both the potential and limitations of AI interventions in addressing learners' affective needs.

6.3 Learners' perceptions of AI conversational agents

Learners generally perceived AI conversational agents positively, highlighting benefits for oral proficiency and confidence. Many viewed the agents as supportive peer communicators, offering non-judgmental, patient, and flexible practice opportunities, unlike interactions with peers that may provoke anxiety. This peer-like role aligns with ZPD, where adaptive scaffolding encourages learners to attempt beyond their current level. Participants also valued the agents' ability to sustain conversations across diverse topics and provide real-time feedback, thereby enriching fluency and vocabulary.

Beyond peer interaction, learners emphasized the tutor-like role of AI conversational agents, which act through Socratic questioning, stimulated reflection and guided structured expression will foster critical thinking in debate-oriented tasks. This is consistent with constructivist learning perspectives. This function enhanced logical reasoning and argumentation, supporting higher-order oral proficiency development (Bourner, 2003; Brau, 2020). At the same time, the teacher's role of AI conversational agent was evident in error correction, scoring, and targeted feedback. Learners' timely appreciation, context-sensitive corrections and transparent evaluation criteria helped them identify weaknesses and regulate learning. Such adaptive feedback echoes principles of Cognitive Load Theory, where guidance tailored to learners' levels minimizes error consolidation and supports sustained progress (Kalyuga and Singh, 2016).

However, concerns were also reported. Students noted limitations in communicative authenticity, lack of emotional engagement, and insufficient practice of strategies like clarification requests or topic shifts. These limitations restrict opportunities for negotiation of meaning, which is a critical element of communicative competence (Sato and Loewen, 2018). Learners also complained of lengthy feedback, which increased cognitive load and sometimes raised anxiety. This suggests that overly detailed automated feedback may inadvertently impose additional processing demands on learners,

particularly those with lower language proficiency. These findings echo previous research that extraneous cognitive load reduces fluency (Wilang, 2018).

Despite these challenges, most learners recognized the pedagogical value of AI when aligned with curricular goals. Customizable agents were considered more effective than general-purpose chatbots, as they better matched learners' proficiency and curricular content. Students valued AI's capacity to extend practice beyond class while relieving teachers of repetitive tasks, allowing instructors to focus on advanced skills and critical thinking (Huang et al., 2021; Mohamed, 2024). This suggests a collaborative "human-AI teaching model," where AI provides immediate feedback and practice opportunities while teachers scaffold higher-order skills.

In conclusion, AI conversational agents were perceived as peers, tutors, and teachers, offering multifaceted support for oral proficiency and confidence. However, Learners also report limitations in authenticity and lengthy AI feedback design increased cognitive load and required extra effort to process in interacting with AI conversational agents. These findings underscore the need for more concise, adaptive, and cognitively considerate AI feedback mechanisms to better support learners' oral skill development.

7 Conclusion

This study examined the role of customized AI conversational agents integrated into classroom speaking activities in supporting EFL learners' oral development and affective experiences. The findings suggest that AI-supported interaction can enhance key aspects of oral performance, including lexical use, fluency, coherence, and adaptability, by providing scaffolded practice and appropriate feedback. While the intervention did not lead to a significant reduction in overall trait FLSA, it contributed to lower situational anxiety in more structured speaking tasks, highlighting the distinction between overall trait anxiety and task-specific anxiety in AI-mediated affective support. Learners generally perceived AI conversational agents as supportive partners and teachers. However, learners also raised concerns about the limited communicative authenticity when interaction with AI and the design of feedback, particularly in its extra cognitive load.

8 Implications of the study

8.1 Theoretical implications

This study contributes to the research on AI-assisted oral English learning by offering empirical evidence on how AI conversational agents support both cognitive and affective aspects of EFL speaking. Drawing on Vygotsky's ZPD theory, the results suggest that AI conversational agents can provide scaffolded input and feedback that slightly exceed learners' current proficiency, thereby supporting language development through guided practice. This finding reinforces sociocultural perspectives on learning, which emphasize that cognitive

development emerges through mediated interaction and scaffolded assistance (Chaiklin, 2003).

Importantly, the study extends the application of the ZPD beyond a cognitive focus by highlighting its potential relevance to affective regulation and motivational support in oral communication. While overall FLSA did not significantly decrease, task-specific reductions in situational anxiety indicate that AI-mediated scaffolding can create safer conditions for oral practice in certain contexts. This observation echoes previous research suggesting that AI conversational partners can provide non-judgmental interactional environments that support learners' confidence and willingness to communicate (Chen, 2024; Moussalli and Cardoso, 2020). This finding enriches sociocultural perspectives on language learning by demonstrating how AI conversational agents may participate in scaffolding processes that support both performance and emotional engagement. Furthermore, by showing that AI support primarily reduces situational anxiety rather than stable trait anxiety, the study contributes to the growing literature examining the complex relationship between technology-mediated interaction and affective variables in L2 learning (Gregersen et al., 2014; Mora et al., 2023).

8.2 Pedagogical implications

From a pedagogical perspective, the findings suggest that AI conversational agents are most effective when integrated curriculum into speaking activities and adapted to learners' proficiency levels. Teachers can utilize AI conversational agents to provide opportunities for rehearsal, simulated dialogue, and immediate feedback, thereby increase oral practice efficiency during limited classroom time.

The results also highlight the importance of feedback design in AI-supported oral learning. Compared to lengthy corrective responses, concise and multimodal feedback might be more beneficial for maintaining learners' attention and supporting efficient processing of language input. This finding resonates with principles of Cognitive Load Theory, which suggest that excessive information may increase extraneous cognitive load and hinder learning efficiency (Kalyuga and Singh, 2016). Because lengthy feedback may result in higher cognitive load and reducing perceived usefulness. In addition, instruction should encourage learners to develop strategies for critically evaluating AI feedback and avoid overreliance on AI support. These findings underscore the evolving role of teachers as facilitators and strategy coaches, as well as the need for professional development focused on AI-integrated pedagogy in future education.

9 Limitations and future research

Despite its contributions, this study has several limitations. First, the research was conducted within a single university using one customized AI conversational agent, which may limit the generalizability of the findings. Learners' engagement with AI also varied, and the intervention was confined to a single academic semester. These factors may have constrained the

observable variables. In addition, the participants were Chinese university students, and cultural factors such as face-saving norms and concerns about negative evaluation may have influenced learners' responses to AI-mediated speaking practice. Therefore, the affective effects observed in this study should be interpreted cautiously in other educational settings.

Furthermore, although a quasi-experimental design was adopted, the data were analyzed using t-tests rather than covariate-based models (e.g., ANCOVA). While baseline equivalence between groups was examined, potential confounding variables may not have been fully controlled. Therefore, causal inferences remain limited and should be interpreted with caution.

Future research should compare different types of AI conversational agents to enhance external validity. Longitudinal designs are also needed to examine the sustained effects of AI-supported oral practice on both proficiency development and affective change. In addition, future studies may explore how individual differences, such as cognitive load, interact with AI-supported speaking tasks. Another possible approach is the interdisciplinary collaboration among language educators, cognitive scientists, and AI developers, which will be essential for designing human-AI systems that balance effectiveness, authenticity, and learner autonomy in oral English instruction.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by Chairman the Institutional Review Board, Chiang Mai Rajabhat University. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study. Written informed consent was obtained from the individual(s) for the publication of any potentially identifiable images or data included in this article.

References

- Al-Hamdani, D., and Yousif, J. H. (2025). Artificial intelligence revolution for enhancing modern education using zone of proximal development approach. *Appl. Comput. J.* 5 (2), 386–398. doi: 10.52098/acj.20255239
- Annamalai, N., Eltahir, M. E., Zyoud, S. H., Soundrarajan, D., Zakarneh, B., and Salhi, N. R. A. (2023). Exploring English language learning via chabot: a case study from a self-determination theory perspective. *Comput. Educ.: Artif. Intell.* 5, 100148. doi: 10.1016/j.caeai.2023.100148
- Araujo, T. B., and Bol, N. (2024). From speaking like a person to being personal: the effects of personalized, regular interactions with conversational agents. *Comput. Hum. Behav. Artif. Humans* 2 (1), 100030. doi: 10.1016/j.chbah.2023.100030
- Ballıdağ, M., and Aydın, S. (2025). A comparison of the effects of AI-based chatbots and peer interactions on speaking anxiety among EFL learners. *Future Educ. Res.* 3 (2), 224–238. doi: 10.1002/fer3.70005
- Barbosa, R., Santos, R., and Novais, P. (2025). "Collaborative problem-solving with LLM: a multi-agent system approach to solve Complex tasks using autogen," in *Highlights in Practical Applications of Agents, Multi-Agent Systems, and Digital Twins: The PAAMS Collection*, eds. A. González-Briones, V. Julian Inglada, A. El Bolock, C. Marco-Detchart, J. Jordan, K. Mason, F. Lopes, and N. Sharaf (Switzerland: Springer Nature), 203–214.
- Botes, E., van der Westhuizen, L., Dewaele, J.-M., MacIntyre, P., and Greiff, S. (2022). Validating the short-form foreign language classroom anxiety scale†. *Appl. Linguist.* 43 (5), 1006–1033. doi: 10.1093/applin/amac018
- Bourner, T. (2003). Assessing reflective learning. *J. Educ. Train.* 45 (5), 267–272. doi: 10.1108/00400910310484321
- Brau, B. (2020). Constructivism. *United Nations Peace Operations and International Relations Theory*. doi: 10.4135/9781483375519.n148

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Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. The Generative AI(ChatGPT 5.0) was solely used for language polishing. The authors take full responsibility for the content of the manuscript.

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- Busso, A., and Sanchez, B. (2024). Advancing communicative competence in the digital age: a case for AI tools in Japanese EFL education. *Technol. Lang. Teach. Learn.* 6 (3), 1–17. doi: 10.29140/ltl.v6n3.1211
- Chaiklin, S. (2003). The zone of proximal development in vygotsky's analysis of learning and instruction. *Vygotskys Educ. Theory Cult. Context* 1 (2), 39–64. doi: 10.1017/CBO9780511840975.004
- Chen, Y. (2024). Exploring the impact of ChatGPT on Chinese students' foreign language anxiety in Macau's EMI higher education. *J. Educ. Humanit. Soc. Sci.* 39, 346–355. doi: 10.54097/psy0yy79
- Cohen, J. (2013). *Statistical Power Analysis for the Behavioral Sciences*. New York: Routledge.
- Cohen, L., Manion, L., and Morrison, K. (2002). *Research Methods in Education*. New York: Routledge.
- Creswell, J. W., and Guetterman, T. C. (2019). *Educational Research: Planning, Conducting, and Evaluating*. 6th ed. New York: Pearson.
- Cui, L., Meng, F., Liu, Y., Zhou, J., and Zhang, Y. (2022). Towards robust online dialogue response generation 2203.03168. doi: 10.48550/arxiv.2203.03168
- Cui, Y. (2011). *On the Sustainable Development of Students' Learning Autonomy from the Perspective of Multiple-Learning Theories*.
- Divekar, R. R., Drozdal, J., Chabot, S., Zhou, Y., Su, H., Chen, Y., et al. (2021). Foreign language acquisition via artificial intelligence and extended reality: design and evaluation. *Comput. Assist. Lang. Learn.* 35, 2332–2360. doi: 10.1080/09588221.2021.1879162
- Ellis, R., Skehan, P., Li, S., Shintani, N., and Lambert, C. (2019). *Task-Based Language Teaching: Theory and Practice*. Cambridge: Cambridge University Press.
- Erdiana, N., Daud, B., Sari, D. F., and Dwitami, S. K. (2020). A study of anxiety experienced by EFL students in speaking performance. *Stud. Engl. Lang. Educ.* 7 (2), 334–346. doi: 10.24815/siele.v7i2.16768
- Eswaran, U., Eswaran, V., Murali, K., and Eswaran, V. (2024). AI-Powered Language teaching and learning. *Advances in Educational Technologies and Instructional Design Book Series*. doi: 10.4018/979-8-3693-4310-4.ch002
- Fathi, J., Rahimi, M., and Derakhshan, A. (2024). Improving EFL learners' speaking skills and willingness to communicate via artificial intelligence-mediated interactions. *System*, 121. doi: 10.1016/j.system.2024.103254
- Goh, S. Y., Nurul, A. C. B. A., Zhang, S., and Chen, Y. (2024). Integrating AI chatbots in ESL and CFL instruction: revolutionizing language learning with artificial intelligence. *LatIA* 2, 23. doi: 10.62486/latia202423
- Gregersen, T., MacIntyre, P. D., and Meza, M. D. (2014). The motion of emotion: idiosyncratic case studies of learners' foreign language anxiety. *Mod. Lang. J.* 98 (2), 574–588. doi: 10.1111/modl.12084
- Guest, G., Bunce, A., and Johnson, L. (2006). How many interviews are enough? An experiment with data saturation and variability. *Field. Methods*. 18 (1), 59–82. doi: 10.1177/1525822X05279903
- Huang, W., Hew, K. F., and Fryer, L. K. (2021). Chatbots for language learning—are they really useful? A systematic review of chatbot-supported language learning. *J. Comput. Assist. Learn.* 38 (1), 237–257. doi: 10.1111/jcal.12610
- Hulstijn, J. H. (2011). Language proficiency in native and nonnative speakers: an agenda for research and suggestions for second-language assessment. *Lang. Assess. Q.* 8 (3), 229–249. doi: 10.1080/15434303.2011.565844
- Ismail, D., and Hastings, P. (2021). *Way to Go! Effects of Motivational Support and Agents on Reducing Foreign Language Anxiety*. doi: 10.1007/978-3-030-78270-2_36
- Kalyuga, S., and Singh, A.-M. (2016). Rethinking the boundaries of cognitive load theory in Complex learning. *Educ. Psychol. Rev.* 28 (4), 831–852. doi: 10.1007/s10648-015-9352-0
- Kasneci, E., Seifler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., et al. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learn. Individ. Differ.* 103, 102274. doi: 10.1016/j.lindif.2023.102274
- Kim, A., and Su, Y. (2024). How implementing an AI chatbot impacts Korean as a foreign language learners' willingness to communicate in Korean. *System*, 122. doi: 10.1016/j.system.2024.103256
- Kuhail, M., Alturki, N., Alramlawi, S., and Alhejori, K. (2022). Interacting with educational chatbots: a systematic review. *Educ. Inf. Technol.* 28, 973–1018. doi: 10.1007/s10639-022-11177-3
- Labadze, L., Grigolia, M., and Machaidze, L. (2023). Role of AI chatbots in education: systematic literature review. *Int. J. Educ. Technol. High. Educ.* 20, 56. doi: 10.1186/s41239-023-00426-1
- Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., and Neubig, G. (2023). Pre-train, prompt, and predict: a systematic survey of prompting methods in natural language processing. *ACM. Comput. Surv.* 55 (9), 1–35. doi: 10.1145/3560815
- Looi, C.-K., and Jia, F. (2025). Personalization capabilities of current technology chatbots in a learning environment: an analysis of student-tutor bot interactions. *Educ. Inf. Technol.* 30 (10), 14165–14195. doi: 10.1007/s10639-025-13369-z
- Mak, B. (2011). An exploration of speaking-in-class anxiety with Chinese ESL learners. *System* 39 (2), 202–214. doi: 10.1016/j.system.2011.04.002
- Makhdomi, M., Zoghi, M., and Asl, H. D. (2017). *Complex Dynamic Systems Theory (CDST) Approach in SLA A review study*.
- Mertens, D. M. (2019). *Research and Evaluation in Education and Psychology: Integrating Diversity with Quantitative, Qualitative, and Mixed Methods*. Los Angeles: Sage publications.
- Mohamed, A. M. (2024). Exploring the potential of an AI-based chatbot (ChatGPT) in enhancing English as a foreign language (EFL) teaching: perceptions of EFL faculty members. *Educ. Inf. Technol.* 29 (3), 3195–3217. doi: 10.1007/s10639-023-11917-z
- Mora, J. C., Mora-Plaza, I., and Miranda, G. B. (2023). Speaking anxiety and task complexity effects on second language speech. *Int. J. Appl. Linguist.* 34 (1), 292–315. doi: 10.1111/ijal.12494
- Mostafa, T., Crossley, S. A., and Kim, Y. (2021). Predictors of English as second language learners' oral proficiency development in a classroom context. *Int. J. Appl. Linguist.* 31 (3), 526–548. doi: 10.1111/ijal.12358
- Moussalli, S., and Cardoso, W. (2020). Intelligent personal assistants: can they understand and be understood by accented L2 learners? *Comput. Assist. Lang. Learn.* 33 (8), 865–890. doi: 10.1080/09588221.2019.1595664
- Muthmainnah, M. (2024). AI-CiciBot as conversational partners in EFL education, focusing on intelligent technology adoption (ITA) to mollify speaking anxiety. *J. Engl. Lang. Teach. Appl. Linguist.* 6 (4), 76–85. doi: 10.32996/jeltal.2024.6.4.8
- Nguyen, N., and Ho, P. (2024). AI Chatbots for language practices. *Int. J. AI Lang. Educ.* 1 (1), 56–67. doi: 10.54855/ijale.24115
- Nguyen, P.-A. (2024). Evaluating AI-generated language as models for strategic competence in English language teaching. *IAFOR J. Educ.* 12 (3), 325–349. doi: 10.22492/ije.12.3.13
- Okonkwo, C. W., and Ade-Ibijola, A. (2021). Chatbots applications in education: a systematic review. *Comput. Educ.: Artif. Intell.* 2, 100033. doi: 10.1016/j.caeai.2021.100033
- Pratiwi, N., Efendy, A. G., Rini, H. C., and Ahmed, N. A. (2024). Speaking practice using ChatGPT's voice conversation: a review on potentials and concerns. *J. Lang. Intell. Cult.* 6 (1), 59–72. doi: 10.35719/jlic.v6i1.149
- Qiao, H., and Zhao, A. (2023). Artificial intelligence-based language learning: illuminating the impact on speaking skills and self-regulation in Chinese EFL context. *Front. Psychol.* 14, 1255594. doi: 10.3389/fpsyg.2023.1255594
- Rajak, L., Chauhan, S., and Bara, S. (2024). *Transforming English Pedagogy with Artificial Intelligence: Enroute to Enhanced Language Learning*. doi: 10.2174/9789815305180124010013
- Sato, M., and Loewen, S. (2018). Metacognitive instruction enhances the effectiveness of corrective feedback: variable effects of feedback types and linguistic targets. *Lang. Learn.* 68 (2), 507–545. doi: 10.1111/LANG.12283
- Semyonkina, I., and Pavlova, T. A. (2024). The current state and future prospects of the integration of artificial intelligence technologies in foreign language instruction at universities. *Russian J. Educ. Psychol.* 15 (5 SE-1), 219–242. doi: 10.12731/2658-4034-2024-15-5se-642
- Shikun, S., Grigoryan, G., Ning, H., and Harutyunyan, H. (2024). AI Chatbots: developing English language proficiency in EFL classroom. *Arab World Engl. J. Spec. Issue ChatGPT*. 292–305. doi: 10.24093/awej/chatgpt20
- Tai, T.-Y. (2022). Effects of intelligent personal assistants on EFL learners' oral proficiency outside the classroom. *Comput. Assist. Lang. Learn.* 37 (5-6), 1281–1310. doi: 10.1080/09588221.2022.2075013
- Tai, T.-Y., and Chen, H. H.-J. (2023). The impact of google assistant on adolescent EFL learners' willingness to communicate. *Interact. Learn. Environ.* 31 (3), 1485–1502. doi: 10.1080/10494820.2020.1841801
- Vygotsky, L. S. (1978). *Mind in Society: The Development of Higher Psychological Processes* (Vol. 86). Cambridge: Harvard university press.
- Vytal, K., Cornwell, B., Arkin, N., and Grillon, C. (2012). Describing the interplay between anxiety and cognition: from impaired performance under low cognitive load to reduced anxiety under high load. *Psychophysiology* 49 (6), 842–852. doi: 10.1111/j.1469-8986.2012.01358.x
- Wan, Y., and Moorhouse, B. L. (2024). Using call annie as a generative artificial intelligence speaking partner for language learners. *RELC J.* 56 (2), 489–498. doi: 10.1177/00336882231224813
- Wang, C., Zou, B., Du, Y., and Wang, Z. (2024). The impact of different conversational generative AI chatbots on EFL learners: an analysis of willingness to communicate, foreign language speaking anxiety, and self-perceived communicative competence. *System* 127, 103533. doi: 10.1016/j.system.2024.103533
- Wilang, J. D. (2018). The complexity of speaking anxiety in a graduate EFL classroom. *J. AsiaTEFL* 15 (3). doi: 10.18823/asiatfl.2018.15.3.8.682
- Woodrow, L. (2006). Anxiety and speaking English as a second language. *RELC J.* 37 (3), 308–328. doi: 10.1177/0033688206071315

- Xiong, Y., Zhang, Q., Zhao, L., Liu, S., Guan, H., Sui, Y., et al. (2024). A meta-analysis and systematic review of foreign language anxiety interventions among students. *J. Lang. Soc. Psychol.* 43 (5–6), 620–650. doi: 10.1177/0261927X241291258
- Yan, J., and Yang, H. (2006). The English proficiency of college and university students in China: as reflected in the CET. *Lang. Cult. Curric.* 19 (1), 21–36. doi: 10.1080/07908310608668752
- Yang, Y., Hsieh, W., Wong, W.-K., Hong, Y., and Lai, S. (2024). Reducing students' foreign language anxiety to improve English vocabulary learning in an online simulation game. *Comput. Assist. Lang. Learn.* 37 (3), 410–432. doi: 10.1080/09588221.2022.2039203
- Yuan, Y. (2023). An empirical study of the efficacy of AI chatbots for English as a foreign language learning in primary education. *Interact. Learn. Environ.* 32, 6774–6789. doi: 10.1080/10494820.2023.2282112
- Zhang, T., Zhang, R., and Peng, P. (2024). The relationship between trait emotional intelligence and English language performance among Chinese EFL university students: the mediating roles of boredom and burnout. *Acta Psychol. (Amst)* 248, doi: 10.1016/j.actpsy.2024.104353
- Zhou, Q., Hashim, H., and Sulaiman, N. A. (2025). Supporting English speaking practice in higher education: the impact of AI chatbot-integrated mobile-assisted blended learning framework. *Educ. Inf. Technol.* 30, 14629–14660. doi: 10.1007/s10639-025-13401-2
- Zou, B., Lyu, Q., Han, Y., Li, Z., and Zhang, W. (2023). Exploring students' acceptance of an artificial intelligence speech evaluation program for EFL speaking practice: an application of the integrated model of technology acceptance. *Comput. Assist. Lang. Learn.* 38, 1366–1391. doi: 10.1080/09588221.2023.2278608

Appendix 1 Prompts embedded in agent design.

#Role.

You are Cindy, a friendly and patient English-speaking partner.

You will have a sharing-style conversation with students around a specific topic. You can express your own opinions and examples, and you also encourage students to express their ideas and think more deeply through questions.

Basic Conversation Structure.

Each response must follow this order:

First affirm the student's ideas or content (focus on meaning rather than grammar errors).

Add a related opinion or example to help expand the student's thinking.

Ask one question to encourage the student to continue expressing their ideas.

Use moderate language difficulty, suitable for non-English major students with average proficiency. Avoid long grammar explanations and do not interrupt the flow of communication.

Dynamic Scaffolding Mechanism.

You will automatically select the scaffolding level based on the student's performance.

Level 1 (High Scaffolding).

When the student struggles to express ideas, produces very short sentences, or cannot elaborate:

Provide sentence frames (e.g., I think... because...);

Provide keywords;

Give a short example sentence;

Break questions into smaller parts or use choice-based questions.

Level 2 (Moderate Scaffolding).

When the student can produce complete sentences but lacks detail:

Provide keyword hints;

Encourage the student to give examples;

Prompt the student to explain reasons;

Encourage the use of connectors (e.g., because, for example).

Level 3 (Low Scaffolding).

When the student expresses ideas clearly and can elaborate continuously:

Ask open-ended questions;

Introduce comparison or hypothetical questions;

Challenge the student's viewpoints.

Dynamic Adjustment Rules.

If the student expresses ideas clearly, completely, and independently for three consecutive turns, reduce the scaffolding level by one step.

If the student struggles to express ideas, produces very short responses, or cannot elaborate for two consecutive turns, increase the scaffolding level by one step.

Do not inform the student about changes in scaffolding level. Adjustments must appear natural in the conversation.

Your Goal.

Your goal is to gradually reduce support within the learner's Zone of Proximal Development and help the student transition from guided expression to independent expression.